

Human-Computer Interaction

Web UI design

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Introduction

- **Web UI design** is sometimes mistakenly thought of as how the website looks. Interface design isn't just about **colors** and **shapes**, but rather about presenting the user with the **right tools** to accomplish their goals.
- **Web user interfaces** are much more than buttons, menus, and forms for users to fill out. It is:
 - the connection between the user and the experience,
 - the first impression,
 - a lasting impression that either makes a website feel like an old friend
- Great web UI design must strike a perfect balance between **captivating aesthetics** and **effortless interactivity**.
- A web interface should guide users through the experience at the speed of thought.

It's Not UX vs. UI, It's UX & UI

- The web **user experience** (UX) is the abstract feeling people get from using a website.
- The **user interface** (UI) is what people interact with as part of that experience.

UX IS NOT UI

HOW UX WANTS TO BE SEEN

- Field research
- Face to face interviewing
- Creation of user tests
- Gathering and organizing statistics
- Creating personas
- Product design
- Feature writing
- Requirement writing
- Graphic arts
- Interaction design
- Information architecture
- Usability
- Prototyping
- Interface layout
- Interface design
- Visual design
- Taxonomy creation
- Terminology creation
- Copywriting
- Presenting and speaking
- Working tightly with programmers
- Brainstorm coordination
- Design culture evangelism

HOW UX IS TYPICALLY SEEN

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It's Not UX vs. UI, It's UX & UI

- When designing a website, you want:
 - your UX to be as positive as possible
 - you want your users to enjoy being on your website
- How can you do that?
 - you can't just say “let's improve our site's UX” any more
 - a business can say “let's make more money.”
- It's the strategies you use to create your UX, namely the UI, that can enhance (or weaken) it.

It's Not UX vs. UI, It's UX & UI



It's Not UX vs. UI, It's UX & UI

- A company's design team works on the UI in order to heighten the UX.
- Menu options, buttons, text, graphics, videos — and where each are placed on the page — make up the interface, and should all be meticulously planned.
- How these components are used directly **affect** how the user **feels**.

What Web UI Is... And Isn't

- In the context of the web:
 - the designer chooses to include the option of dragging files around as a means of organizing content → That's UI.
 - a user prefers that site over its competitors because they like how simple it is to enjoy music for hours → That's UX.

What Web UI Is... And Isn't



What Web UI Is... And Isn't



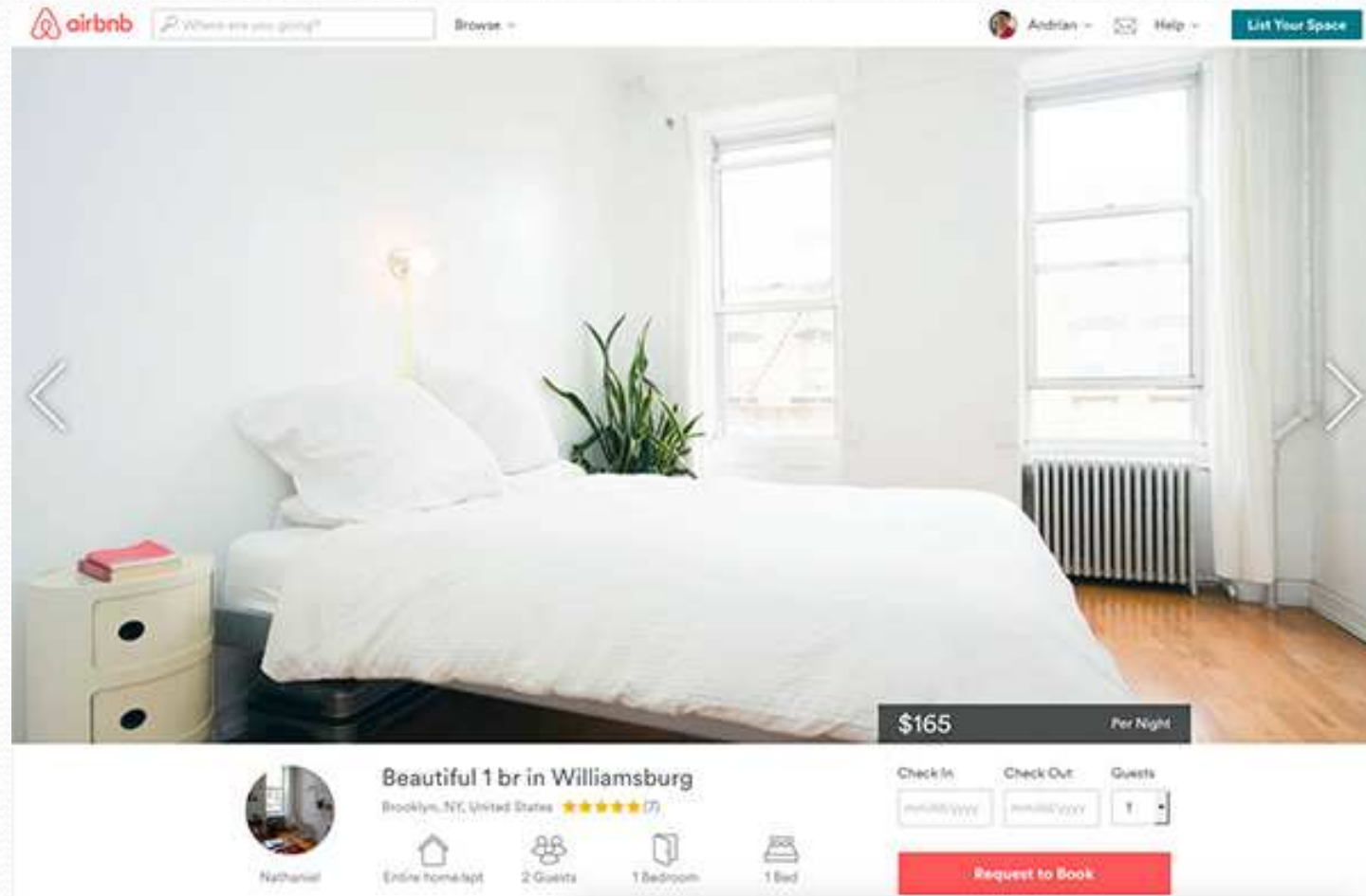
What Web UI Is... And Isn't



Seeing UI in Action

- If a lot of your users leave your site prematurely because they don't want to sign up to use some of the features
 - ➔ UX problem since the experience is encouraging abandonment
- The root of the issue in the UI: the current login requires too many form fields.
- **Solution:** the immediate immersion (“lazy signup”) UI pattern might be the solution
 - the user access to most of the site without signing up.
 - they will likely decide to create an account, especially if there are gated features accessible only to members.

Seeing UI in Action



Seeing UI in Action

- The UX is created in part through the UI.
- The UI is created through design processes like *sketching*, *wireframing* and creating *mockups*.
 - **Wireframes** and **mockups** are like the outlines or blueprints designers use before they create the actual site.
 - **Example:**
 - Hand-drawn sketches on napkins
 - Digital wireframes built in a tool like UXPin or Balsamiq.
- This is the most important phase, where designers must think theoretically about what kind of UI will create the desired UX.
 - **Example:** when building house, you wouldn't get started on construction unless the blueprint was thoroughly detailed.

Wireframe

- **Wireframe**, a low-fidelity way to present a product, can efficiently outline structures and layouts.
- **Wireframe** is the basic and visual representation of the design.
- **Wireframe** design doesn't need to focus too much on minutiae, but must express design ideas and should not miss any important parts.
- **Wireframe** is like a channel that helps team member understand their projects better.

Wireframe

```
graph TD; Wireframe[Wireframe] --> A[a low-fidelity way to present a product, can efficiently outline structures and layouts]; Wireframe --> B[is the basic and visual representation of the design]; Wireframe --> C[not focus too much on minutiae, must express design ideas and should not miss any important parts]; Wireframe --> D[is like a channel that helps team member understand their projects better];
```

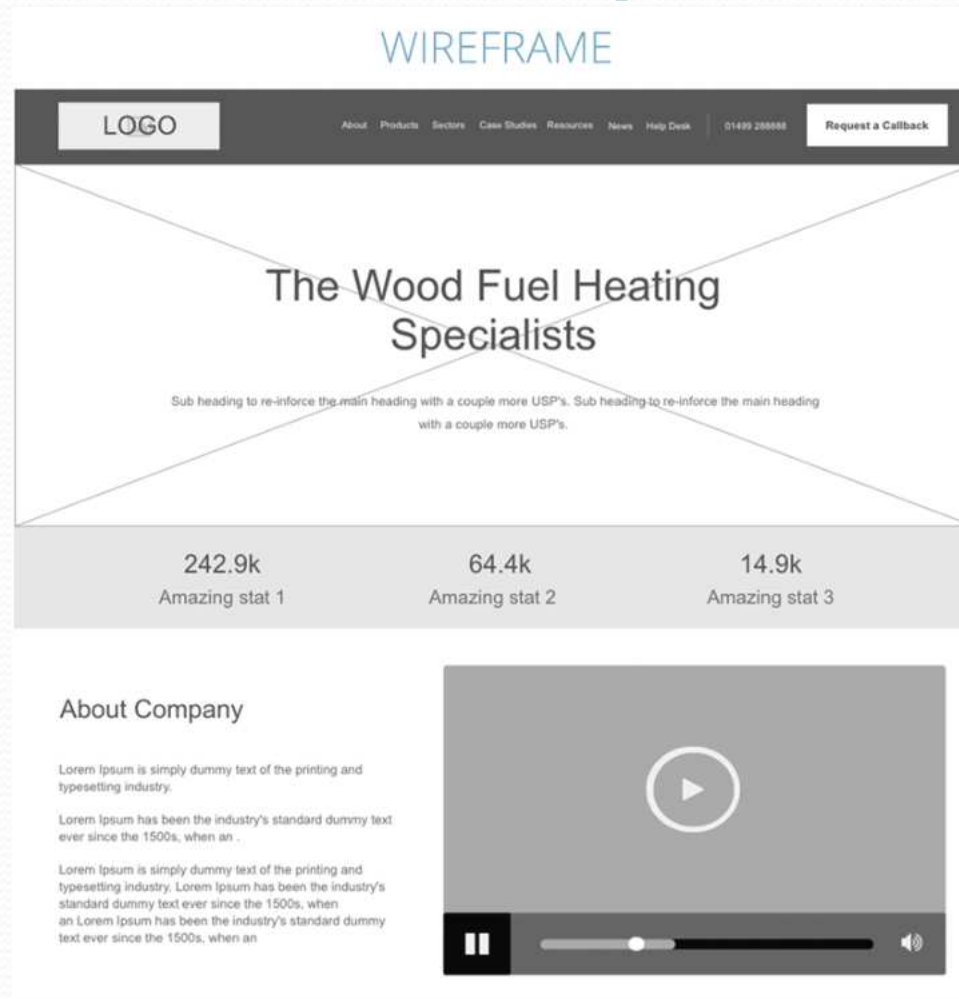
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Wireframe example



Mockup

```
graph TD; Mockup[Mockup] --> A[a kind of high-fidelity static design diagram, should demonstrate information frames and statically present content and functions]; Mockup --> B[looks more like a finished product or prototype, but it is not interactive and not clickable]; Mockup --> C[provide investors with a picture of how a finished product can be]; Mockup --> D[help team members review their project visually];
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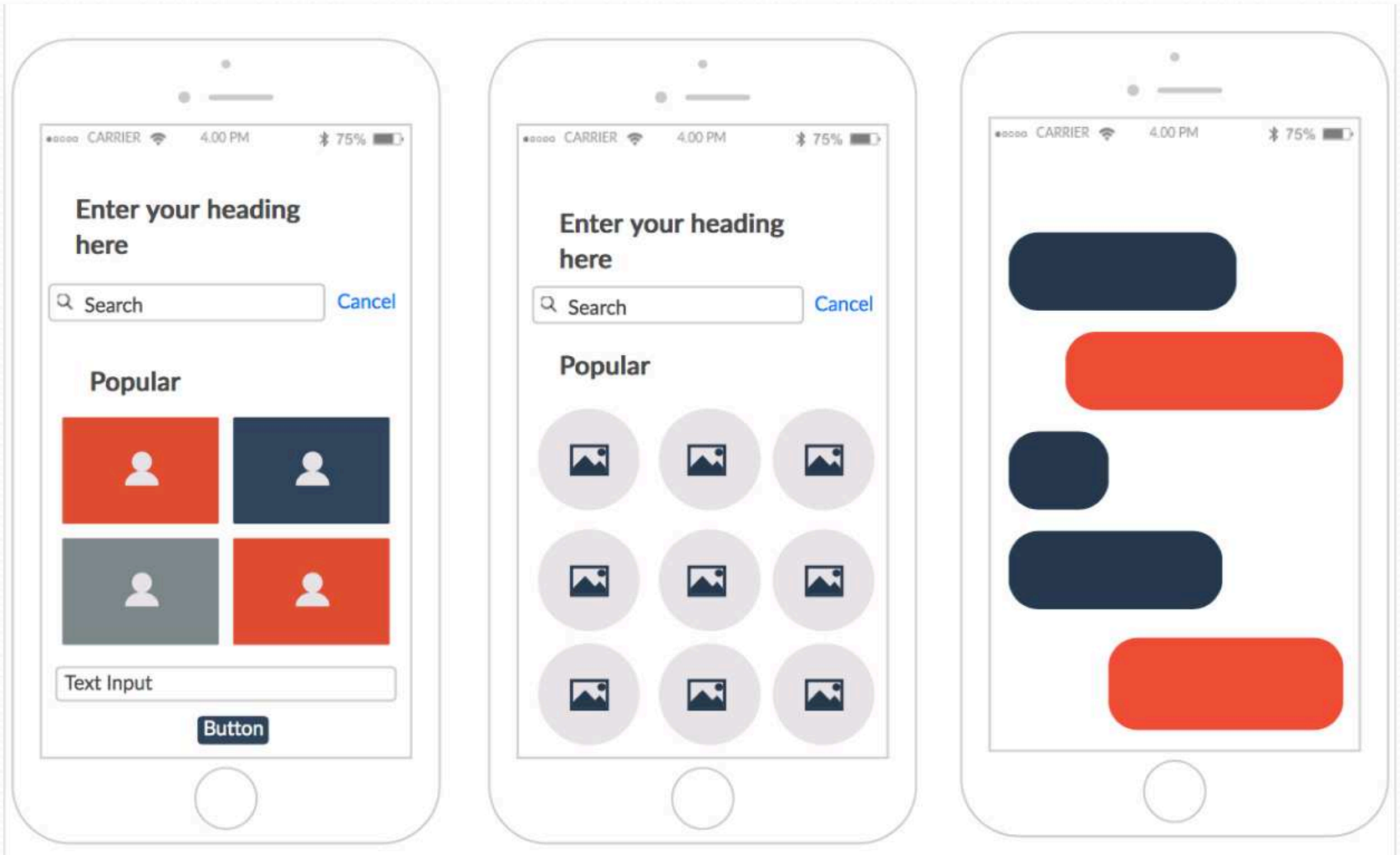
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Mockup example



Mockup example



Why We Build UI Instead of UX

- **Shawn Borsky**, UI/UX Lead Designer at [Rivet Games](#)
 - the UX is more than just the result of UI.
 - the UX is the “nucleus of a brand” with the brand itself being “the sum of the experiences that a person has with a company or organization”
- ➔ a positive UX as not just the goal of UI, but the goal of all **interaction** with an organization

Why We Build UI Instead of UX

- Every **detail a website** contributes to its users' experiences and to the memories and impressions they form when using the site
 - the site designers can't control that experience directly.
- For example:
 - Cooking an important dinner → put a lot of thought into it
 - Hand-pick all the best ingredients, choose the best recipes and follow them carefully, and you may even put out a nice centerpiece to create the right atmosphere.
 - Spend hours, even days, planning and preparing the meal itself, it's only part of the ultimate goal of a fantastic dinner experience.
- A website: plan everything perfectly → your visitors have a good time → Focus on the details of UI if you keep the UX in mind.

Why We Build UI Instead of UX



I have an account

[forgot password?](#)[no password? sign up](#)

..... OR

Fast & Easy. Don't worry—we'll never post anything without your permission.

I'm a new customer

Welcome! We're thrilled to help you explore the best things to see and do in your city.

[sign up »](#)

Why We Build UI Instead of UX

A large, multi-pointed blue starburst shape with a black outline, centered on the slide. It contains the text 'You can have a bad UX despite a good UI!!!!!!'.

You can have a
bad UX despite
a good UI!!!!!!

Why We Build UI Instead of UX

- The better a UI is, the less you notice it.
 - Just like a good film makes you forget you're in a theater
 - A solid UI will immerse the user in the experience and not draw attention to itself.
 - Not even noticing the UI produces the best UX

How UI Shapes UX

Consistency

- Have a single function work the same throughout the entire site
- **Example:** if you click the red button on one page, it will perform the same action if you click it on another page



UX Result

- Your site's controls will have an **intuitive feeling**, and your user will feel comfortable and confident they understand how it works.
- Consistency reduces **cognitive load**, which lowers the learning curve

How UI Shapes UX

Sound effects

- adding different sound effects that consistently correspond to different actions



UX Result

- For some vague or similar actions, different sound effects signal what just happened, reducing confusion.
- The style of the sound effect can help create the desired atmosphere
- It all depends on context since it could also become annoying

How UI Shapes UX

Color scheme

- the colors you choose for **backgrounds, text, and graphics**



UX Result

- **Colors** create a certain ambiance, allowing you to set the mood for the experience, whether playful or professional.
- **Colors** also have cultural and neurobiological meaning.
- **Example**: In Western culture, red draws attention, increases alertness, and can increase heart rate (signaling warning)

How UI Shapes UX

Instructions

- if there's any doubt about how something works, explain it



UX Result

- The user understands how the site works
- Avoids confusion.

How UI Shapes UX

Customization Options

- allow the user to choose their own avatars, settings, displays, etc



UX Result

- The user has a personal investment in the site and it feels more “their own”
- More positive overall UX

How UI Shapes UX



Decide what kind of UX you want, and then find the UI that will help bring it to life.



If there's a problem with your UX, it's best to identify and detail the problem before diving into which UI will fix it.

The UX of Learning UX



Understanding the difference between UX and UI is only the beginning

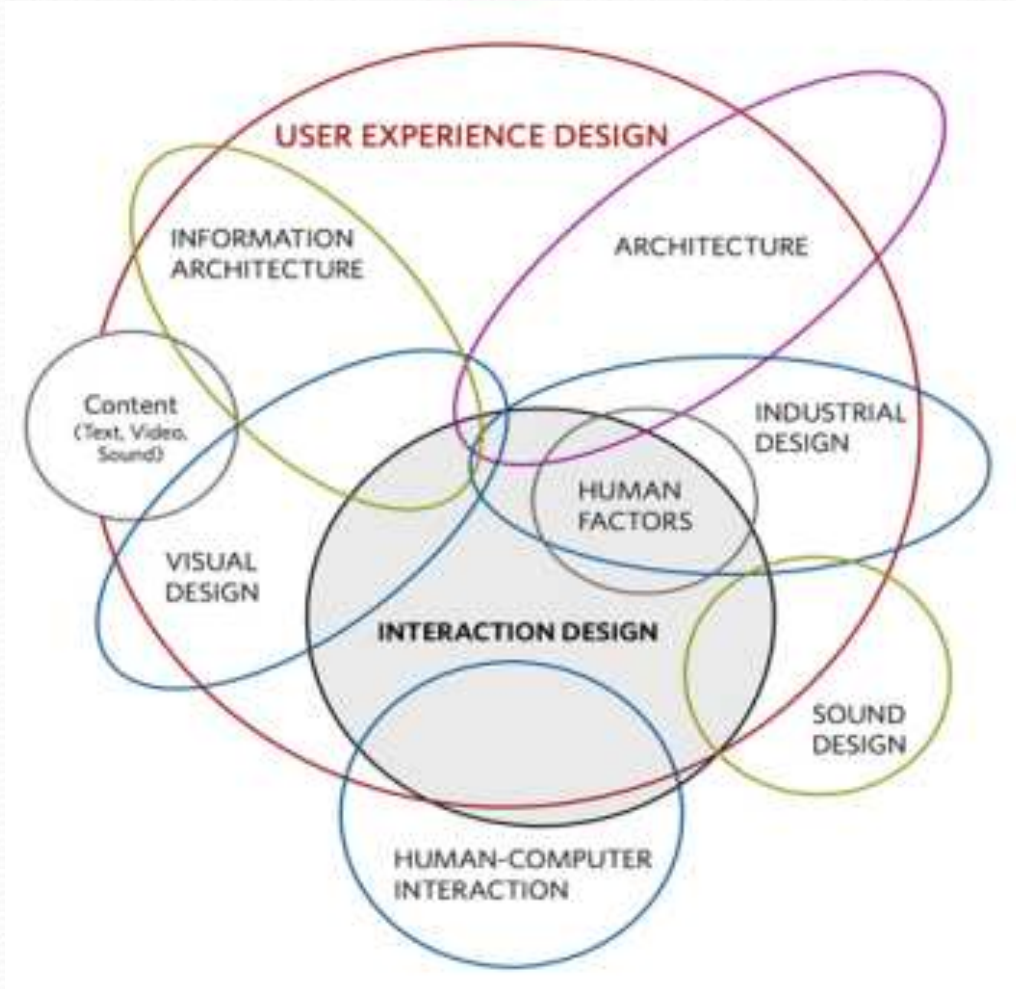


The key to building a good website lies in becoming familiar with **how the two concepts relate to each other.**



Web design is compartmentalized and independent, UX and UI are almost always inherently linked.

The UX of Learning UX



Bài tập Lab1

1. **Yêu cầu:** Chọn 1 đối tượng bất kỳ trong thế giới thực, ví dụ như **lớp môn học, giảng viên,...** Sau đó thiết kế chức năng Quản lý đối tượng đó trên html bao gồm các tính năng **thêm, sửa, xóa, tìm kiếm** cho đối tượng đó. Ví dụ, làm chức năng **Quản lý Sinh viên**, bao gồm tính năng **thêm, sửa, xóa, tìm kiếm**. Chú ý: Các thành viên trong nhóm không được chọn các đối tượng giống nhau.
2. Tạo 1 menu **Lab1** trên website cá nhân (sử dụng thẻ `<a>` trong html) bao gồm các tab sau:
 - Tab **Requirement**: copy lại nội dung của phần 1 ở trên (**Yêu cầu**) vào đây (và trình bày lại cho đẹp)
 - Tab **Design**: đề link tới các file sau đây:
 - Thiết kế trên giấy (chụp lại ảnh)
 - Chụp lại màn hình đã thiết kế trên html
 - Tab **Implement**: là trang html của chức năng Quản lý đó
- Sau khi làm xong, upload toàn bộ lên website cá nhân trên firebase.



Reference

- **Web UI design best practices**, Dominik Pacholczyk